

CoRoT-2b

R-1357

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-2_190830 · created 2026-07-31T23:12:26+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2458725.7491 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.203 +/- 0.014
Transit depth (Rp/Rs)^2	4.12 +/- 0.55 [%]
Orbital Inclination (inc)	85.12 +/- 1.37
Airmass coefficient 1 (a1)	0.975 +/- 0.019
Airmass coefficient 2 (a2)	0.0145 +/- 0.0094
Scatter in the residuals of the lightcurve fit is	1.96 %
Best Comparison Star	#3 - [201, 39]
Optimal Aperture	4.37
Optimal Annulus	11.19
Transit Duration (day)	0.0868 +/- 0.0075

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.203 ± 0.014 vs 0.1667 (deviation 2.46σ)
Duration fit vs published	0.0868 d vs 0.095 d (deviation 1.04σ)
Mid-time O–C	-6.7 min
Depth SNR	13.8
Residual scatter	1.96 %

Light curve

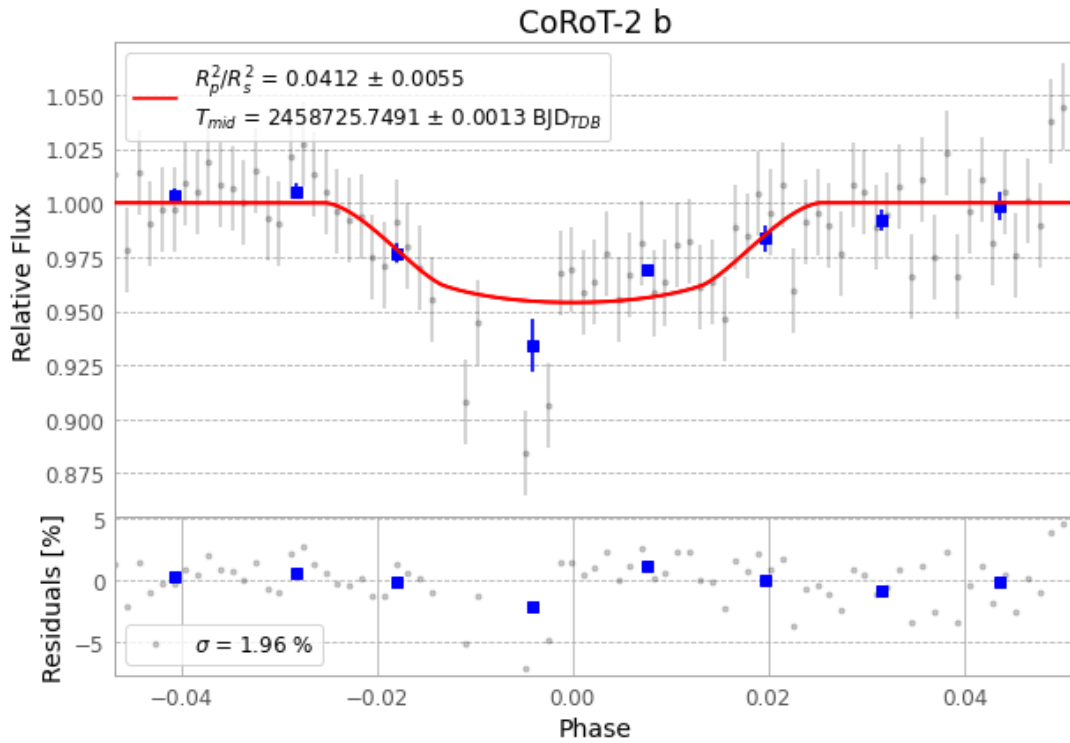


Figure 1 — FinalLightCurve_CoRoT-2 b_2019-08-29.png (SHA-256 e92f62dbf423026c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [160, 225], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d2d4ea1567000857...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

83 FITS frames (55 MB), CoRoT-2190830035412.FITS → CoRoT-2190830080012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-190830.log (d73b0e6ed625...), FinalLightCurve_CoRoT-2 b_2019-08-29.png (e92f62dbf423...), FinalLightCurve_CoRoT-2 b_2019-08-29.csv (bee214dfbc1c...)

Compute resources

Wall time 155.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 23:12Z.